# Cardiorespiratory Emergence:
# Heart and Lung as Coupled Relaxation Oscillators under Reflex Control

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**Living version.** Canonical, retrieval-ready HTML: `jamming-physics.org/cardioresp_vp_site/` · Concept DOI `10.5281/zenodo.20755371` · Reproduction code: `github.com/rego093-sketch/jamming-physics`.

**Abstract**

The heart and the lung are *emerged* here, in the jamming-physics idiom, from the measured stacking stabilities ($\gamma$) of their master genes — NKX2-5 for the heart, NKX2-1 for the lung — read from the 4D-DNA gene-clock and never refitted. That single number sets each organ's $R_{19}$ bistable switch, whose cubic nonlinearity is exactly the fast law of a FitzHugh–Nagumo (FHN) relaxation oscillator; the heartbeat and the breath are therefore two settings of *one* DNA-anchored oscillator, separated only by the slow recovery time $\tau_s$ (heart $\tau_s = 18$, lung $\tau_s = 95$). From a single cited time anchor (resting heart rhythm, 1.17 Hz = 70 bpm) the model derives the eupnea rate (16.86 breaths/min, breath-to-beat ratio 0.240), the apnea / periodic-breathing threshold (a delayed-chemoreflex Hopf bifurcation at $\mathrm{LG}/\mathrm{LG}_c = 1$), respiratory sinus arrhythmia (a high-frequency heart-rate-variability peak locking to the respiratory frequency 0.2811 Hz), the Cheyne–Stokes period (period $\approx 2\times$ circulatory delay; slope 1.992), and baroreflex regulation ($-1.0$ bpm/mmHg to a 70 bpm set-point). A shared $R_{19}$ bistable-switch kernel yields a carcinogen dose–response with $\mathrm{RR}(0) = 1$, no threshold, and a convex shape, reaching $\approx 18.3\times$ at 100 pack-years. Every displayed quantity carries an honest grade ([F] forced, [V] verified, [L] literature-anchored, [O] open), and every number is regenerated by a deterministic engine (two-run identical SHA-256, whole-harness drift 0). This is a grounded, DNA-derived study of cardiorespiratory *mechanism dynamics*; it makes no clinical, pharmacological, diagnostic, or treatment claim.

## 1 Scope and firewall

This package models cardiorespiratory *mechanism dynamics*: relaxation oscillators, delayed control loops, and bistable switches. The governing firewall is that $\gamma$ is a promoter stacking-stability read that fixes organ *identity* and developmental *order* (cited from the DNA gene-clock), and is *never* equated with a clinical magnitude — a heart rate, a channel voltage, a drug potency, a dose, or an incidence rate. Every *absolute* physiological or epidemiological magnitude is graded [O] and listed with its obstacle (§12); the *shapes and orderings* that follow from the substrate are forced [F] or verified [V].

The result is not a toy curve fit. No constant is chosen to hit a target: the master-gene $\gamma$ values are measured DNA inputs, the time anchor is fixed once by a single cited resting rate, and every other rate is a model rate divided by that one anchor. Because the inputs are fixed by DNA, the dynamics are falsifiable — different measured $\gamma$ would move the recovery-time ratio, the breathing rate, and the carcinogen low-dose slope together, in the predicted direction.

## 2 From DNA: the emergence chain

The dynamics descend from DNA through one chain: **gene** $\to \gamma \to R_{19}$ **switch** $\to$ **oscillator**. Each organ begins at a master gene whose promoter stacking stability $\gamma$ is read from the 4D-DNA gene-clock; that one number fixes the $R_{19}$ bistable switch (barrier $B = \gamma^2/4$, spinodal $h^* = 2(\gamma/3)^{3/2}$), and the switch's cubic is the fast nonlinearity of the FHN oscillator. Ordering the two organs by ascending $\gamma$ returns the developmental order lung $\to$ heart, owned by the gene-clock and read out here exactly.

| stage | heart | lung | grade |
|---|---|---|---|
| master gene | NKX2-5 | NKX2-1 | cited (DNA) |
| stacking stability $\gamma$ | 1.513 | 1.5088 | [V] readout |
| functional spinodal | 0.716319 | 0.713338 | [F] substrate |
| $R_{19}$ switch + FHN, $\tau_s$ | 18 | 95 | [F]/[V] |
| functional role | SA-node pacemaker | preBötzinger rhythm | [V] |

The no-tuning rule is what separates a grounded emergence from a fit: one $\gamma$ does all three jobs — it orders the organs, sets the switch barrier, and sets the oscillator. The identity and the order are established results of the 4D-DNA volume; on that DNA-anchored foundation this volume derives the full functional layer that follows.

## 3 The shared relaxation oscillator

Both pacemakers are one FHN unit: a fast excitable variable $s$ with the $R_{19}$ cubic and a slow recovery variable $w$,

$$\dot{s} = \frac{E(\gamma s - s^3) - I\,w + \text{drive}}{\tau_f}, \qquad \dot{w} = \frac{s - \beta w}{\tau_s}. \tag{1}$$

Nothing organ-specific is added beyond $\gamma$ and the slow recovery time $\tau_s$. The slow time sets the rhythm: larger $\tau_s$ gives a slower relaxation cycle, which is precisely why the heart ($\tau_s = 18$) runs fast and the lung ($\tau_s = 95$) runs slow on the same mechanism.

Run free, each unit produces a sustained limit cycle rather than a fixed point; over a fixed window the heart completes 115 cycles and the lung 28, the expected fast/slow split. The oscillator mechanism is verified [V] from the shared substrate; the *absolute* rate is set by the cited anchor of §4, not by the oscillator.

## 4 One time anchor, then eupnea

A single conversion $\kappa$ from model time to seconds is fixed by declaring the heart's intrinsic relaxation rhythm equal to the cited resting heart rate, $1.17\,\text{Hz} = 70\,\text{bpm}$ [L]. That gives $\kappa = 0.016327\,\text{s/arb}$; nothing else about timing is tuned, and every physical rate is the model rate divided by $\kappa$. Under that one anchor the lung's rhythm is fixed by the ratio of recovery times, not by a second fit, giving 16.86 breaths/min — inside the cited eupnea band of 12–18 — with a *structural* breath-to-beat ratio of 0.240 (the $\tau_s$ ratio itself).

| drive | breaths/min | oscillates |
|---|---|---|
| 0.45 | 17.75 | yes |
| 0.50 | 17.34 | yes |
| 0.55 | 16.87 | yes |
| 0.60 | 16.32 | yes |
| 0.65 | 15.67 | yes |
| 0.70 | 14.86 | yes |

Sweeping the excitatory drive keeps the breathing rate inside or adjacent to the eupnea band and never stops the oscillation, so the result is not a fragile operating point. The absolute rate carries the cited anchor's grade [L]; the ratio is verified [V].

## 5 Apnea threshold and chemoreflex loop gain

Periodic breathing (the apnea–hyperpnea cycle) is a delayed negative chemoreflex on a first-order gas plant, $\dot{c} = -(c + V_{\text{dev}})/T_p$ with $V_{\text{dev}} = \text{clamp}\big(\text{LG} \cdot c(t - \tau_d)\big)$. The clamp at zero is the apnea floor: ventilation cannot go negative. Such a loop is stable at low gain and undergoes a Hopf-type instability when the loop gain crosses a critical value; normalised to that value the onset sits at $\text{LG}/\text{LG}_c = 1$.

| $\text{LG}/\text{LG}_c$ | late amplitude | sustained | apnea fraction |
|---|---|---|---|
| 0.5 | 0.00 | no | 0.00 |
| 0.7 | 0.00 | no | 0.00 |
| 0.9 | $\approx 0$ | no | 0.00 |
| 1.1 | 0.94 | yes | 0.26 |

Below threshold the amplitude settles to zero; above it a sustained limit cycle appears with a well-defined period ($\approx 31.74\,\text{s}$ at the example delay). The transition is sharp, with no ambiguous middle band. The existence of the instability and the apnea clamp are verified [V]; the threshold is reported relative to its own critical value, so the qualitative law — loop gain above one drives periodic breathing — is the claim, not a tuned absolute number.

## 6 Respiratory sinus arrhythmia

Respiratory sinus arrhythmia (RSA) is reproduced by letting the respiratory oscillator frequency-modulate the cardiac drive, $\text{drive} = \text{drive}_0 + \varepsilon\, r(t)$; the diagnostic is the heart-rate-variability (HRV) spectrum computed from inter-beat intervals. The lock is defined inside the cited high-frequency HRV window 0.15–0.40 Hz, the standard vagal RSA band [L]. Decoupled, the HRV spectrum shows no in-band peak at the respiratory frequency; coupled, a high-frequency peak appears essentially at $f_{\text{resp}} = 0.2811\,\text{Hz}$ (measured peak 0.2784 Hz) and tracks the respiratory frequency as it is scaled.

| coupling $\varepsilon$ | HRV peak (Hz) | $f_{\text{resp}}$ (Hz) | in-band lock |
|---|---|---|---|
| 0.00 (decoupled) | 0.303 | 0.281 | no |
| 0.02 | 0.284 | 0.281 | yes |
| 0.05 | 0.278 | 0.281 | yes |
| 0.10 | 0.284 | 0.281 | yes |

Coupling creates an HF-HRV peak at the respiratory frequency that is absent when decoupled and that tracks $f_{\text{resp}}$: grade [V]. The band itself is the cited Task-Force window [L].

## 7 Cheyne–Stokes period and circulatory delay

Cheyne–Stokes respiration is the chemoreflex limit cycle in the fast-plant, delay-dominated regime typical of low cardiac output. In that regime the cycle period is set by the loop transport delay, and the model returns a period-versus-delay slope of 1.992 — essentially $T_{\text{CSR}} \approx 2 \times \tau_{\text{circ}}$. This factor of two is the forced relation of the section: a delayed negative feedback oscillates with a period of about twice its loop delay, recovered across a range of delays rather than at a single point.

| circulatory delay (s) | CSR period (s) | period / delay |
|---|---|---|
| 8 | 17.74 | 2.22 |
| 12 | 25.74 | 2.15 |
| 15 | 31.74 | 2.12 |
| 20 | 41.74 | 2.09 |

Because the period scales with the transport delay, a slower circulation (a longer lung-to-chemoreceptor delay) lengthens the cycle; the mechanism predicts the direction directly. The period scaling as about twice the delay is verified [V] across the sweep, with the slope inside the acceptance band $[1.7, 2.4]$.

## 8 Baroreflex correction of blood pressure

The baroreflex is modelled at the rate level, because a relaxation oscillator's cycle rate is correctly insensitive to its drive — the autonomic loop must set the SA-node operating rate, not the FHN drive. A blood-pressure disturbance drives a compensating heart-rate command with cited gain and latency: the static sensitivity is $-1.0$ bpm/mmHg with the correct sign (a pressure rise lowers heart rate), and the onset latency is 0.68 s, both at cited values [L].

| $\Delta P$ (mmHg) | $\Delta$HR (bpm) | slope | latency (s) | sign |
|---|---|---|---|---|
| $-20$ | $+20$ | $-1.0$ | 0.68 | correct |
| $-10$ | $+10$ | $-1.0$ | 0.68 | correct |
| $+10$ | $-10$ | $-1.0$ | 0.68 | correct |
| $+20$ | $-20$ | $-1.0$ | 0.68 | correct |

The response is linear over $\pm 20\,\mathrm{mmHg}$ and the closed loop returns pressure to its set-point, with heart rate settling at the cited resting 70 bpm. Gain and latency are literature-anchored [L]; the verified content is the correct sign, the linearity, and the closed-loop regulation.

## 9 Carcinogen dose–response (shared $R_{19}$ kernel)

Cell fate is the same $R_{19}$ bistable switch. A carcinogen is a sustained aberrant bias $h_c$ that pushes toward the malignant basin and lowers the barrier the normal basin must climb; the malignant-crossing rate is Kramers/Arrhenius over that barrier, rate $\propto \exp(-B_{\mathrm{eff}}/\mathrm{scale})$, and relative risk is $\mathrm{RR}(\mathrm{dose}) = \mathrm{rate}(\mathrm{dose})/\mathrm{rate}(0)$. The barrier is derived, not postulated: the three critical points are the real roots of $s^3 - \gamma s - h = 0$ (three reals inside $|h| < h^* = 2(\gamma/3)^{3/2}$), and $B_{\mathrm{eff}} = U(\mathrm{saddle}) - U(\mathrm{basin})$. Two exact identities follow and are checked at import: $B_{\mathrm{eff}}(h{=}0) = \gamma^2/4 = 0.569119$ [F], and the barrier-drop slope at the origin equals $\sqrt{\gamma} = 1.2283$ [F].

Mapping cumulative tobacco exposure to bias gives a lung-carcinoma relative risk that starts exactly at $\mathrm{RR}(0) = 1$, rises monotonically, and is convex, reaching $18.3\times$ at 100 pack-years — the Doll & Peto order of magnitude. There is no threshold: even half a pack-year already gives $\mathrm{RR} = 1.021 > 1$. The low-dose slope is the forced quantity $\sqrt{\gamma}$, so any positive exposure raises risk — the linear-no-threshold shape, grade [V].

| pack-years | relative risk | ln RR |
|---:|---:|---:|
| 0 | 1.000× | 0.000 |
| 5 | 1.228× | 0.205 |
| 10 | 1.499× | 0.405 |
| 20 | 2.198× | 0.788 |
| 30 | 3.151× | 1.148 |
| 40 | 4.411× | 1.484 |
| 50 | 6.026× | 1.796 |
| 60 | 8.019× | 2.082 |
| 80 | 13.019× | 2.566 |
| 100 | 18.319× | 2.908 |

Two carcinogens add their bias $h$; the model then predicts a combined risk that is $0.76\times$ the product of the singles — sub-multiplicative at moderate dose, approaching multiplicative only in the low-dose limit. This is a falsifiable corollary; empirical agent-pair synergy varies. The *absolute* relative-risk magnitude is open [O]: the switch's effective noise scale and the molecular dose-to-bias conversion are not fixed by substrate geometry and need an external population-hazard calibration — exactly as absolute organ size is open while size order is forced. The forced endpoints ($\mathrm{RR}(0) = 1$, no threshold) and the convex shape stand independently of that calibration.

## 10 Shared-substrate cardiorespiratory spectrum

The three primitives above — the $R_{19}$ switch, the FHN oscillator, and the delayed chemoreflex loop — reproduce a spectrum of cardiorespiratory phenomena with no new mechanism added. These are dynamical signatures of the shared substrate, stated as mechanism, not as clinical, diagnostic, or management statements.

| phenomenon | reused primitive | verified signature | open |
|---|---|---|---|
| Mayer waves $\approx 0.1\,\mathrm{Hz}$ | baroreflex loop (§9) | resonance exists [V] | freq. cited [L] |
| airway-tone switch | $R_{19}$ reversible bias (§9) | bistable + hysteresis [V] | provocation [O] |
| obstructive-apnea regime | loop gain (§6) + patency switch | loop-gain endotype [V] | collapse pressure [O] |
| periodic-breathing spectrum | chemoreflex Hopf (§6, §8) | directions + delay scaling [V] | thresholds [O] |
| fever co-scaling | shared oscillator (§3) | HR–RR co-scaling [V] | slope [O] |
| excitable cough | FHN excitable regime (§3) | all-or-none firing [V] | threshold [O] |

Slow arterial-pressure waves near 0.1 Hz emerge as a resonance of the same delayed baroreflex loop; reversible airway-tone collapse is the $R_{19}$ switch run as a reversible (rather than one-way) bias; obstructive sleep apnea reads as a pharyngeal-patency switch on top of the ventilatory loop-gain instability; altitude, low-output, and opioid periodic breathing form one spectrum of the chemoreflex Hopf instability at different delays and gains; because both organs are one oscillator a thermal speed-up raises the cardiac and respiratory rate together; and cough is the FHN unit in its excitable regime, where a sub-threshold stimulus decays and a supra-threshold stimulus fires a full response. Across the spectrum the verified content is the mechanism reuse; every *absolute* magnitude is grade [O] with the obstacle recorded in §12.

## 11 Methods, determinism, and the reproducibility apparatus

Every number is regenerated by one deterministic engine: pinned single-thread linear algebra, a fixed seed, round-before-hash, and sorted-key output. A two-run check yields an identical SHA-256 (engine hash begins `25900a90079a`; whole-harness result blob `413554bcae98`). A single command, `python repro/run_all.py`, emerges the organs from $\gamma$, confirms the shared oscillator, runs the five-target stress battery and the carcinogen kernel, and then runs eleven reproducibility gates ending in one line: `OVERALL: PASS (11/11), drift 0`. This apparatus was inherited in full from the analgesic-threshold VP volume (`10.5281/zenodo.20733420`).

| gate | checks |
|---|---|
| R1 engine determinism | two engine runs give an identical SHA-256 |
| R2 report drift-0 | the whole result blob is byte-identical across recomputation |
| R3 stress battery | the five T1–T5 discriminants pass |
| R4 oncology kernel | $\mathrm{RR}(0) = 1$, no threshold, barrier matches the substrate |
| R5 shared-substrate spectrum | every reuse holds with no new primitive |
| R6 file manifest | every shipped file matches a SHA-256 manifest (fail-closed) |
| R7 forbidden-claim scan | no drug, dosing, or treatment language on the site (fail-closed) |
| R8 grade legality | only the four legal grade tokens appear |
| R9 open-item ledger | every open item on the site states an obstacle |
| R10 concept DOI | the DOI is embedded site-wide with no placeholder |
| R11 offline | the engine reproduces with the network disabled |

Beyond the engine output, a file-level SHA-256 manifest (51 entries) pins the whole delivered package, so any silent byte-drift in a shipped file is caught. The build carries no wall-clock — the release date is pinned — so regenerating the canonical site reproduces it byte-for-byte, and the engine reproduces with the network socket disabled.

## 12 Grades and the irreproducibility ledger

Four grades, and no others, appear: [F] forced by the $R_{19}$ substrate, [V] reproduced and numerically checked, [L] anchored to a cited literature value, and [O] open with a stated obstacle. The package has exactly three open items, each tracing to the same un-calibrated switch noise scale and stimulus-to-bias conversion; their forced/verified counterparts stand independently of the open absolute scale.

| open item | grade | obstacle |
|---|---|---|
| absolute organ size / mass | [O] | DWELL $\propto \gamma^{3/2}$ fixes relative size and order [F]/[V]; absolute magnitude needs an external calibration |
| absolute cancer relative-risk magnitude | [O] | shape ($\mathrm{RR}(0) = 1$, no threshold, convex) is forced/verified; absolute RR needs the switch noise scale and dose$\to$bias conversion (external hazard calibration) |
| absolute disease-spectrum magnitudes | [O] | mechanism reuse verified; absolute provocation, collapse pressure, thresholds, slope, and excitation level need the same external calibration |

## 13 Data and reproduction availability

The reproduction archive is the deposited package `cardioresp_vp_site_v0_5_0.zip` (canonical HTML under `docs/`, the deterministic engine, the stress and oncology and disease modules, the SHA-256 manifest, and the gate harness). Running `python repro/run_all.py` reproduces every displayed number and prints `OVERALL: PASS (11/11), drift 0`. The same code is browsable at `github.com/rego093-sketch/jamming-physics`, the canonical site is `jamming-physics.org/cardiore` and the versioned archive is the concept DOI `10.5281/zenodo.20755371`. All material is released under CC BY 4.0.

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